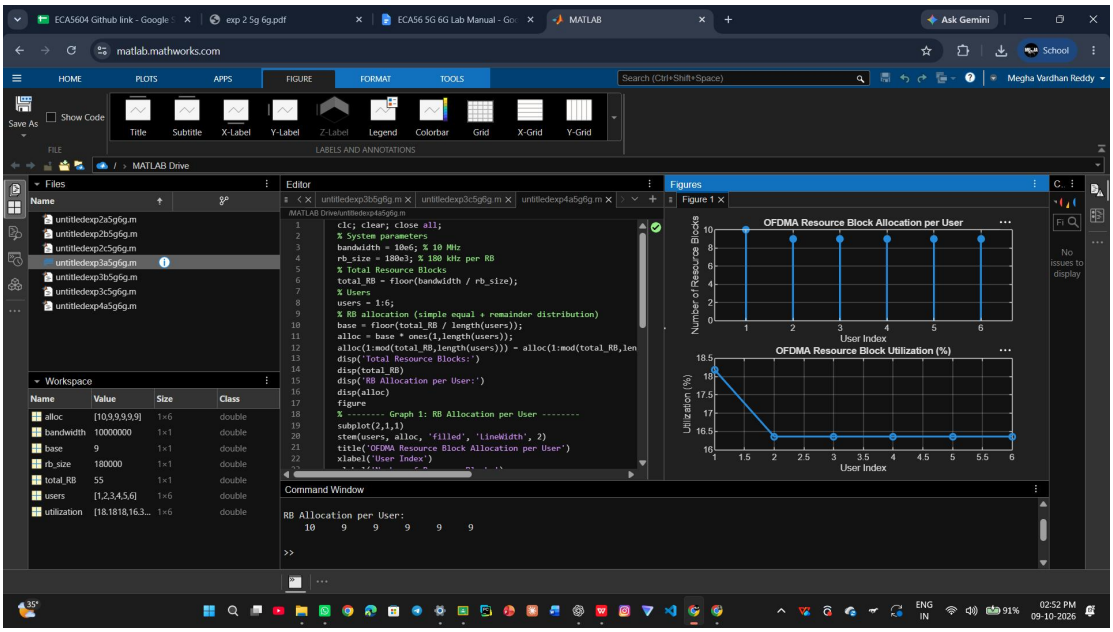
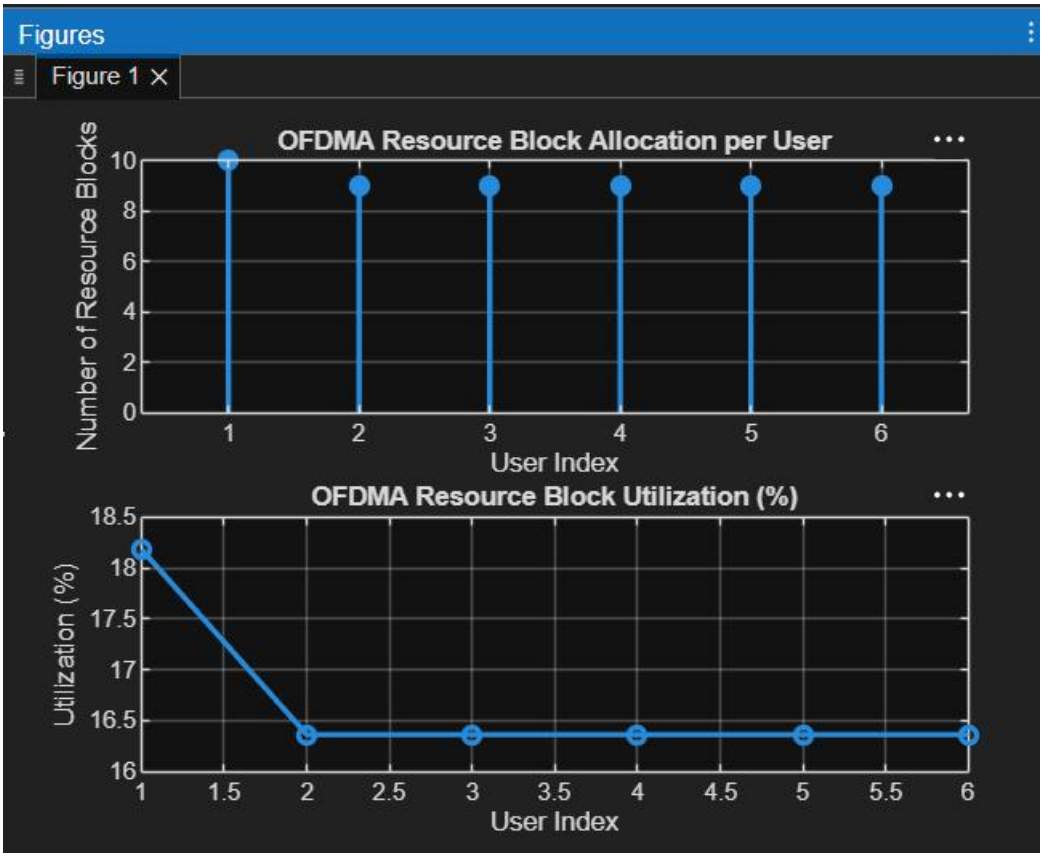
Ex.No.4: OFDMA Resource Block Allocation

Objective

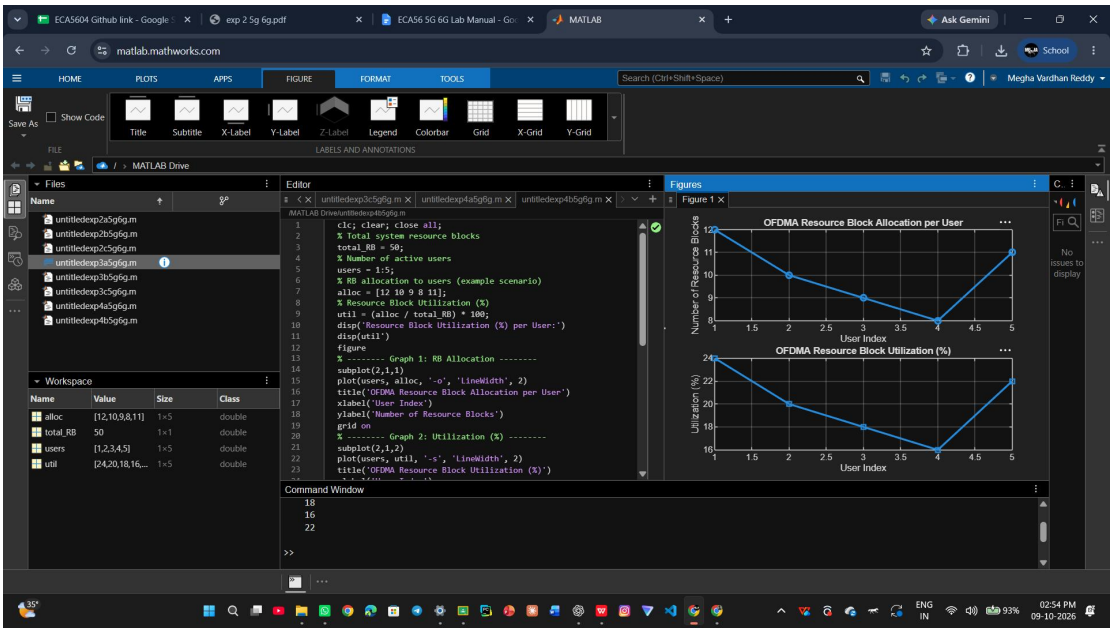
1. To determine the **Number of Resource Blocks (RBs)** allocated in an OFDMA system based on available bandwidth and system configuration.
2. To evaluate the **Resource Block Utilization (%)** by analyzing how efficiently available RBs are assigned to active users.
3. To calculate the **User Throughput (Mbps)** based on allocated resource blocks and modulation efficiency in the OFDMA system.

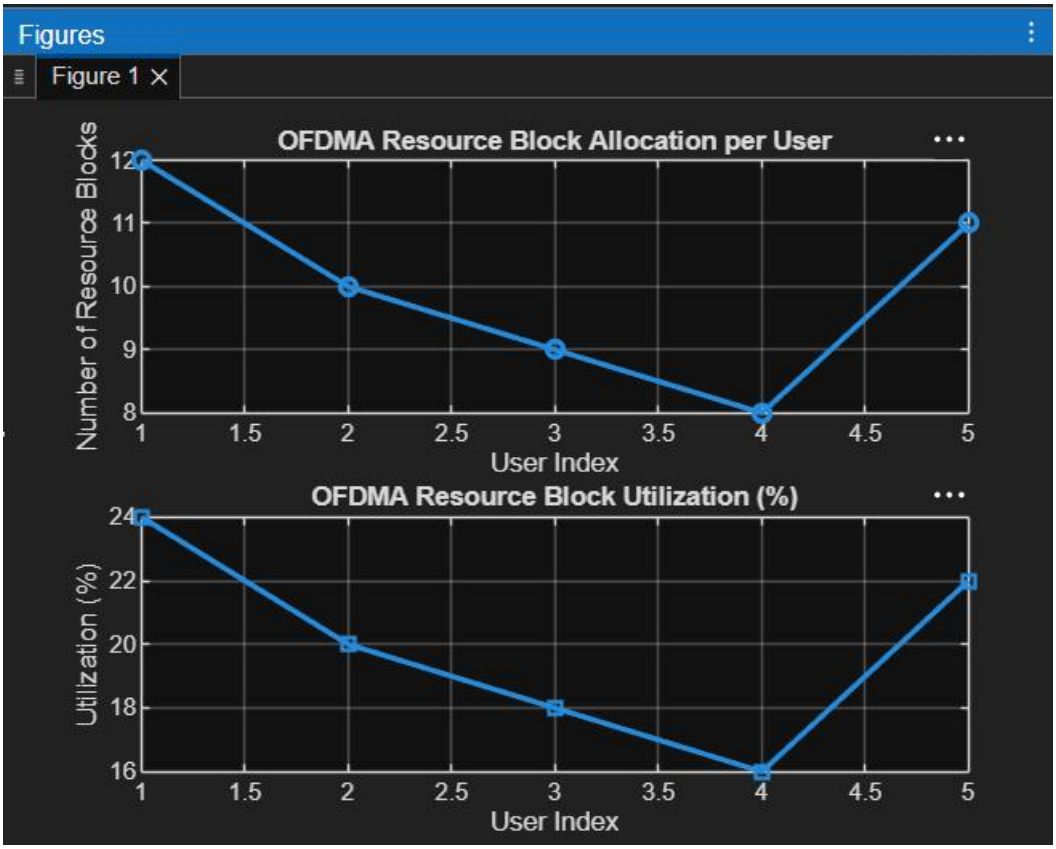
Objective 1 Matlab Code and Output:





Objective 2 Matlab Code and Output:




Figures
Figure 1
OFDMA Resource Block Allocation per User
Number of Resource Blocks
12
11
10
9
8
1
1.5
2
2.5
3
3.5
4
4.5
5
User Index
OFDMA Resource Block Utilization (%)
Utilization (%)
24
22
20
18
16
1
1.5
2
2.5
3
3.5
4
4.5
5
User Index

Objective 3 Matlab Code and Output:

